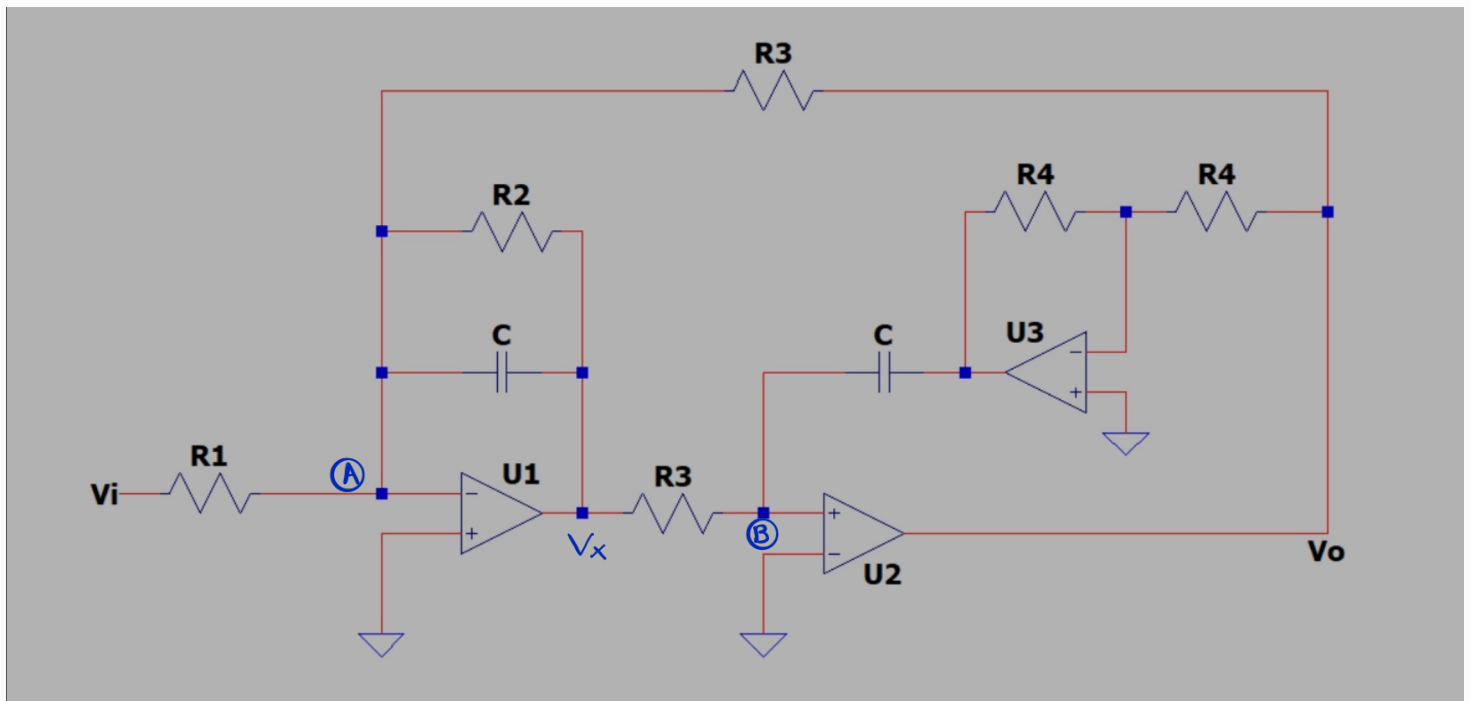




$$\textcircled{B} \quad 0 = -V_x G_3 - (-V_2 sC) \longrightarrow \boxed{V_x = V_2 s \frac{C}{G_3}}$$

$$\textcircled{A} \quad 0 = -V_1 G_1 - V_x sC - V_x G_2 - V_2 G_3$$

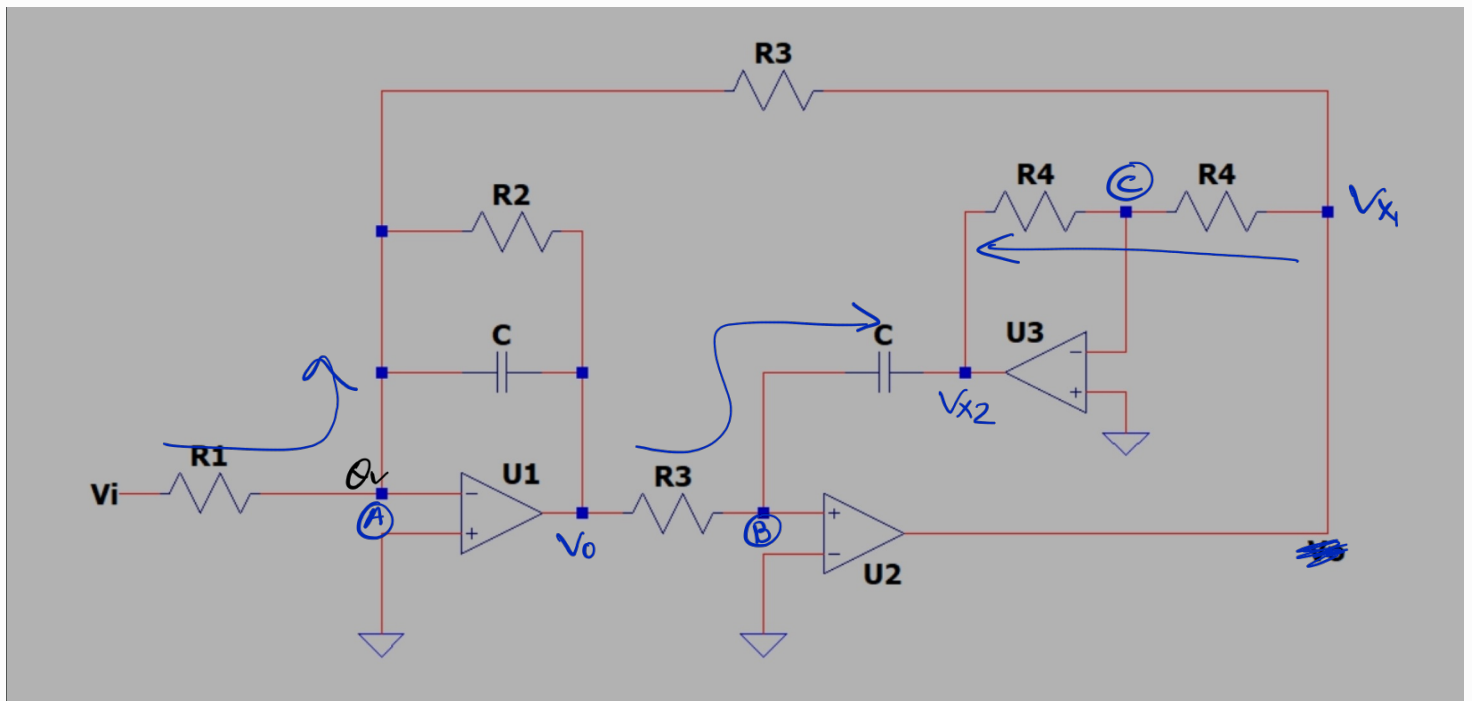
$$\hookrightarrow \frac{V_2}{V_1} = \frac{G_1}{s^2 \frac{C^2}{G_3} + sC \frac{G_2}{G_3} + G_3}$$

$$\frac{V_2}{V_1} = \frac{-\frac{G_1 G_3}{C^2}}{s^2 + s \cdot \frac{G_2}{C} + \left(\frac{G_3}{C}\right)^2}$$

$$\Rightarrow \boxed{T(s, \omega_0, Q) = \frac{K \cdot \omega_0^2}{s^2 + s \frac{\omega_0}{Q} + \omega_0^2}}$$

$$\begin{aligned} K \omega_0 &= -\frac{G_1 G_3}{C^2} \Rightarrow K = -\frac{G_1}{G_3} \\ \frac{\omega_0}{Q} &= \frac{G_2}{C} \rightarrow Q = \omega_0 \frac{C}{G_2} = \frac{G_3}{G_2} \end{aligned} \quad \left. \vphantom{\begin{aligned} K \omega_0 &= -\frac{G_1 G_3}{C^2} \\ \frac{\omega_0}{Q} &= \frac{G_2}{C} \end{aligned}} \right\} \begin{cases} K = -\frac{G_1}{G_3} \\ \omega_0 = \frac{G_3}{C} \\ Q = \frac{G_3}{G_2} \end{cases}$$

$$\boxed{T(s) = \frac{\frac{1}{R_1 R_3 C^2}}{s^2 + s \cdot \frac{1}{R_2 C} + \left(\frac{1}{R_3 C}\right)^2} \cdot \begin{cases} K = -\frac{R_3}{R_1} \\ \omega_0 = \frac{1}{R_3 C} \\ Q = \frac{R_2}{R_3} \end{cases}}$$



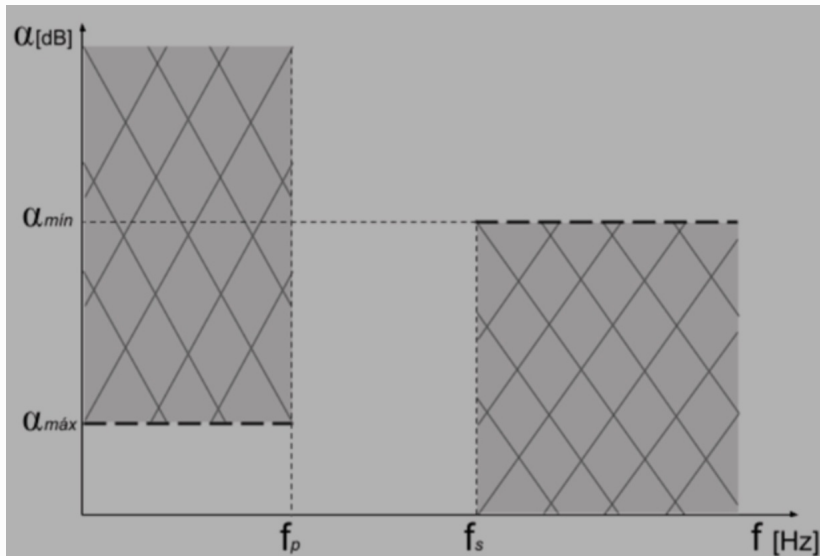
$$\begin{aligned}
 \textcircled{C} \quad V_{x1} G_4 &= -V_{x2} G_4 \rightarrow \boxed{V_{x1} = -V_{x2}} \\
 \textcircled{B} \quad V_0 G_3 &= -V_{x2} \cdot sC \rightarrow \boxed{V_{x2} = -\frac{G_3}{sC} V_0} \\
 \textcircled{A} \quad V_i G_1 &= -(G_2 + sC) V_0 - G_3 V_{x1}
 \end{aligned}$$

$$\rightarrow -\left[(G_2 + sC) + \frac{G_3^2}{sC}\right] V_0 = V_i G_1$$

$$\frac{V_0}{V_i} = \frac{-sC G_1}{sC G_2 + s^2 C^2 + G_3^2} \rightarrow \frac{V_0}{V_i} = \frac{-s \frac{G_1}{C}}{s^2 + s \frac{G_2}{C} + \left(\frac{G_3}{C}\right)^2} //$$

$$\text{Si } T(s) = \frac{K s \frac{\omega_0}{Q}}{s^2 + s \frac{\omega_0}{Q} + \omega_0^2} \Rightarrow \begin{cases} \omega_0 = \frac{G_3}{C} = \frac{1}{R_3 C} \\ Q = \frac{G_3}{C} \cdot \frac{C}{G_2} = \frac{G_3}{G_2} = \frac{R_2}{R_3} \\ K = -\frac{G_1}{C} \cdot \frac{C}{G_2} = -\frac{G_1}{G_2} = -\frac{R_2}{R_1} \end{cases}$$

$$T(s) = \frac{-\frac{R_2}{R_1} \cdot s \frac{1}{R_2 C}}{s^2 + s \frac{1}{R_2 C} + \left(\frac{1}{R_3 C}\right)^2} \quad \omega_0 = \frac{1}{R_3 C}, \quad Q = \frac{R_2}{R_3}, \quad K = -\frac{R_2}{R_1}$$



$\alpha_{m\acute{a}x}$ [dB]	$\alpha_{m\acute{i}n}$ [dB]	f_p [Hz]	f_s [Hz]
0,4	48	3200	9600

$$\alpha(\omega) = 10 \log(1 + \epsilon^2 \cosh^2(n \cdot \cosh^{-1}(\omega)))$$

$$\alpha(1) = \alpha_{m\acute{a}x} = 10 \log(1 + \epsilon^2)$$

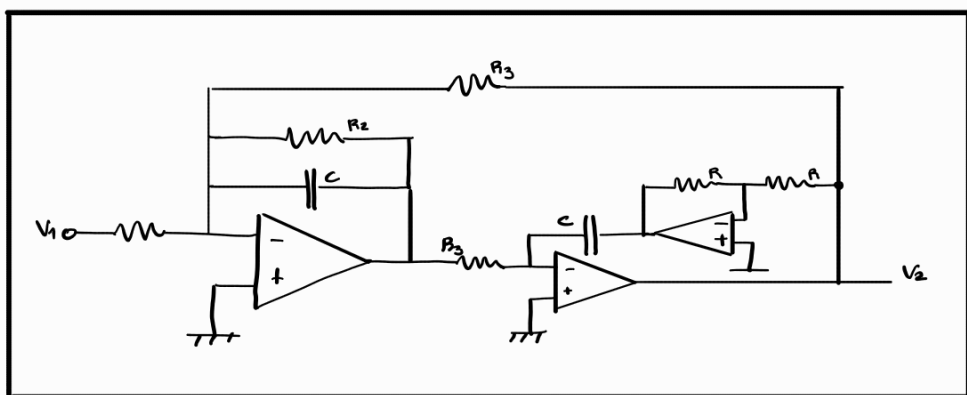
$$\alpha_{m\acute{i}n} = 10 \log(1 + \epsilon^2 \cosh^2(n \cdot \operatorname{arccosh}(\omega_s)))$$

$$\hookrightarrow \boxed{\epsilon^2 = 0,0964}$$

$$\hookrightarrow \epsilon = 0,31$$

n	$\alpha_{m\acute{i}n}$
3	29,76
4	45,06
5	60,37

$$\Rightarrow \boxed{n = 5}$$



$$\boxed{n = 2}$$

